3.2 WONG JIN XUAN

3.2.1 Quarterly and Monthly Borrowing Trend Analysis

SQL Code:

```
SPOOL "C:\Users\wongj\Documents\BorrowingTrend_Report1.txt"

SET LINESIZE 130
SET PAGESIZE 35
SET TRIMSPOOL ON
SET VERIFY OFF
SET SQLBLANKLINES ON
SET WRAP OFF
SET TAB OFF
ALTER SESSION SET NLS_DATE_FORMAT = 'DD-MON-YYYY';

PROMPT === Borrowing Trend Parameters ===
ACCEPT start_year NUMBER DEFAULT 2023 PROMPT 'Enter START year (YYYY) [2023]:
'
ACCEPT end_year   NUMBER DEFAULT 2025 PROMPT 'Enter END year (YYYY)   [2025]:
'

COLUMN run_dt NEW_VALUE run_dt
SELECT TO_CHAR(SYSDATE,'DD-MON-YYYY') run_dt FROM dual;

COLUMN sep_line NEW_VALUE sep_line NOPRINT
SELECT RPAD('_', 130, '_') AS sep_line FROM dual;

TTITLE LEFT ' ' CENTER 'Quarterly and Monthly Borrowing Trend Analysis
&start_year -- &end_year' -
       SKIP 1 RIGHT 'Page ' FORMAT 999 SQL.PNO SKIP 1 LEFT 'Date: &run_dt'
SKIP 2
BTITLE CENTER '&sep_line'

-- ===================== Compact column formats (fit within 130 chars)
=====================
CLEAR COLUMNS
COLUMN "Year"         HEADING 'Year'                FORMAT 9999
COLUMN period_label   HEADING 'Period'              FORMAT A12
COLUMN borrows        HEADING 'Borrows|(count)'     FORMAT 999,999,999
COLUMN last_period    HEADING 'Last|Period'         FORMAT 999,999,999
COLUMN pct_vs_last    HEADING 'Change %|vs Last'    FORMAT 999990.00
COLUMN ma3            HEADING '3-Month|Moving Avg'  FORMAT 999,999,999.99
COLUMN season_var     HEADING 'Seasonal|Index %'    FORMAT A12 JUSTIFY RIGHT
COLUMN net_avail      HEADING 'Net Copies|Available' FORMAT 999,999,999
COLUMN top_genre      HEADING 'Top|Genre'           FORMAT A18
COLUMN festive_events HEADING 'Holiday|Month'       FORMAT A10 JUSTIFY CENTER
```

```
COLUMN yr_group NOPRINT

-- ===================== Breaks & (no computed totals) =====================
CLEAR BREAKS
CLEAR COMPUTES
BREAK ON yr_group SKIP 1 ON REPORT

-- ===================== Hierarchical report =====================
WITH
lib_cap AS (
  SELECT COUNT(*) AS total_copies
  FROM BookCopies
  WHERE UPPER(NVL(bookStatus,'AVAILABLE')) <> 'UNAVAILABLE'
),
base AS (  -- borrowing dates in range
  SELECT d.cal_date,
         EXTRACT(YEAR  FROM d.cal_date) AS yr,
         EXTRACT(MONTH FROM d.cal_date) AS mon
  FROM FactBorrowing fb
  JOIN DimDate d ON d.dateKey = fb.dateKey
  WHERE EXTRACT(YEAR FROM d.cal_date) BETWEEN &start_year AND &end_year
),
m AS (     -- monthly borrow counts
  SELECT yr, mon,
         TRUNC(cal_date,'MM')           AS month_start_dt,
         LAST_DAY(TRUNC(cal_date,'MM')) AS month_end_dt,
         COUNT(*)                       AS borrow_cnt
  FROM base
  GROUP BY yr, mon, TRUNC(cal_date,'MM'), LAST_DAY(TRUNC(cal_date,'MM'))
),
moy AS (  -- average by month-of-year (seasonality baseline)
  SELECT mon AS mon_of_year, AVG(borrow_cnt) AS moy_avg
  FROM m
  GROUP BY mon
),
overall AS (  -- overall mean of monthly averages
  SELECT AVG(moy_avg) AS overall_avg
  FROM moy
),
m_active AS (  -- active loans at month-end
  SELECT m.month_start_dt, COUNT(*) AS active_loans_end
  FROM m
  JOIN BorrowedBooks bb
    ON bb.borrowDate <= m.month_end_dt
   AND (bb.returnDate IS NULL OR bb.returnDate > m.month_end_dt)
  GROUP BY m.month_start_dt
),
```

```
/* Top genre per month (ties kept, joined with '/') */
mg AS (
  SELECT EXTRACT(YEAR  FROM d.cal_date) AS yr,
         EXTRACT(MONTH FROM d.cal_date) AS mon,
         b.genre,
         COUNT(*) AS borrow_cnt
  FROM FactBorrowing fb
  JOIN DimDate d ON d.dateKey = fb.dateKey
  JOIN DimBook b ON b.bookKey = fb.bookKey
  WHERE EXTRACT(YEAR FROM d.cal_date) BETWEEN &start_year AND &end_year
  GROUP BY EXTRACT(YEAR FROM d.cal_date), EXTRACT(MONTH FROM d.cal_date),
b.genre
),
top_gen AS (
  SELECT yr, mon,
         LISTAGG(genre, '/') WITHIN GROUP (ORDER BY genre) AS top_genre
  FROM (
    SELECT yr, mon, genre,
           RANK() OVER (PARTITION BY yr, mon ORDER BY borrow_cnt DESC) AS rnk
    FROM mg
  )
  WHERE rnk = 1
  GROUP BY yr, mon
),
/* NEW: Yes/No holiday flag per (year, month) */
m_hol AS (
  SELECT
    EXTRACT(YEAR  FROM d.cal_date) AS yr,
    EXTRACT(MONTH FROM d.cal_date) AS mon,
    CASE WHEN MAX(CASE WHEN d.holiday_indicator = 'Y' THEN 1 ELSE 0 END) = 1
         THEN 'Yes' ELSE 'No' END AS holiday_month
  FROM DimDate d
  WHERE EXTRACT(YEAR FROM d.cal_date) BETWEEN &start_year AND &end_year
  GROUP BY EXTRACT(YEAR FROM d.cal_date), EXTRACT(MONTH FROM d.cal_date)
),
m_stats AS (  -- monthly stats + seasonality + capacity + top genre + holiday
flag
  SELECT
    m.yr,
    m.mon,
    CEIL(m.mon/3) AS qtr_num,
    MOD(m.mon-1,3)+1 AS mon_in_qtr,
    m.month_start_dt,
    m.month_end_dt,
    m.borrow_cnt,
    LAG(m.borrow_cnt) OVER (PARTITION BY m.yr ORDER BY m.month_start_dt) AS
prev_month,
```

```
    ROUND(
      (m.borrow_cnt - LAG(m.borrow_cnt) OVER (PARTITION BY m.yr ORDER BY
m.month_start_dt))
      / NULLIF(LAG(m.borrow_cnt) OVER (PARTITION BY m.yr ORDER BY
m.month_start_dt),0) * 100, 2
    ) AS pct_vs_last_month,
    ROUND(AVG(m.borrow_cnt) OVER (
            PARTITION BY m.yr ORDER BY m.month_start_dt
            ROWS BETWEEN 2 PRECEDING AND CURRENT ROW), 2) AS ma3,
    ROUND(moy.moy_avg / NULLIF((SELECT overall_avg FROM overall),0) * 100 -
100, 2) AS season_index_pct,
    (SELECT total_copies FROM lib_cap) - NVL(ma.active_loans_end,0) AS
net_avail,
    NVL(tg.top_genre, NULL) AS top_genre,
    h.holiday_month AS festive_events   -- keep printed column name, now
Yes/No
  FROM m
  JOIN moy ON moy.mon_of_year = m.mon
  LEFT JOIN m_active ma ON ma.month_start_dt = m.month_start_dt
  LEFT JOIN top_gen tg ON tg.yr = m.yr AND tg.mon = m.mon
  LEFT JOIN m_hol  h  ON h.yr  = m.yr AND h.mon = m.mon
),
q_end AS (
  SELECT yr, CEIL(mon/3) AS qtr_num, MAX(month_end_dt) AS qtr_end_dt
  FROM m
  GROUP BY yr, CEIL(mon/3)
),
q_active AS (
  SELECT qe.yr, qe.qtr_num, COUNT(*) AS active_loans_end
  FROM q_end qe
  JOIN BorrowedBooks bb
    ON bb.borrowDate <= qe.qtr_end_dt
   AND (bb.returnDate IS NULL OR bb.returnDate > qe.qtr_end_dt)
  GROUP BY qe.yr, qe.qtr_num
),
q_stats AS (  -- quarterly aggregates
  SELECT
    ms.yr,
    ms.qtr_num,
    SUM(ms.borrow_cnt)                                                  AS
qtr_borrow_cnt,
    LAG(SUM(ms.borrow_cnt)) OVER (PARTITION BY ms.yr ORDER BY ms.qtr_num) AS
prev_qtr,
    ROUND(
      (SUM(ms.borrow_cnt) - LAG(SUM(ms.borrow_cnt)) OVER (PARTITION BY ms.yr
ORDER BY ms.qtr_num))
```

```
      / NULLIF(LAG(SUM(ms.borrow_cnt)) OVER (PARTITION BY ms.yr ORDER BY
ms.qtr_num),0) * 100, 2
    ) AS pct_vs_last_qtr,
    (SELECT total_copies FROM lib_cap) - NVL(MAX(qa.active_loans_end),0)   AS
net_avail_qtr
  FROM m_stats ms
  LEFT JOIN q_active qa
    ON qa.yr = ms.yr AND qa.qtr_num = ms.qtr_num
  GROUP BY ms.yr, ms.qtr_num
),
year_tot AS (  -- yearly totals (for footer band)
  SELECT yr, SUM(qtr_borrow_cnt) AS year_borrow_total
  FROM q_stats
  GROUP BY yr
)

SELECT
  ysort                         AS yr_group,   -- hidden break key (NOPRINT)
  yr_display                    AS "Year",
  period_label,
  borrows,
  last_period,
  pct_vs_last,
  ma3,
  season_var,
  net_avail,
  top_genre,
  festive_events
FROM (

  /* Month rows — show Year + Quarter on the FIRST month in each quarter */
  SELECT
    CASE WHEN ms.mon_in_qtr = 1 THEN ms.yr END                 AS yr_display,
-- <== put Year back
    CASE
      WHEN ms.mon_in_qtr = 1 THEN
          RPAD('Q' || ms.qtr_num, 6, ' ')
          || TO_CHAR(ADD_MONTHS(DATE '2000-01-01', ms.mon-1), 'Mon')
      ELSE
          '      ' || TO_CHAR(ADD_MONTHS(DATE '2000-01-01', ms.mon-1), 'Mon')
    END                                                         AS
period_label,
    ms.borrow_cnt                                               AS borrows,
    ms.prev_month                                               AS
last_period,
    ms.pct_vs_last_month                                        AS
pct_vs_last,
```

```
    ms.ma3                                                      AS ma3,
    '    ' || TO_CHAR(ms.season_index_pct, 'SFM990.00') || '%'  AS season_var,
    ms.net_avail                                                AS net_avail,
    ms.top_genre                                                AS top_genre,
    ms.festive_events                                           AS
festive_events,
    ms.yr                                                       AS ysort,
    ms.qtr_num                                                  AS qsort,
    ms.mon_in_qtr                                               AS ord_in_qtr
  FROM m_stats ms

  UNION ALL

  /* Separator BEFORE quarter totals */
  SELECT
    CAST(NULL AS NUMBER)                    AS yr_display,
    RPAD('-',12,'-')                        AS period_label,
    CAST(NULL AS NUMBER)                    AS borrows,
    CAST(NULL AS NUMBER)                    AS last_period,
    CAST(NULL AS NUMBER)                    AS pct_vs_last,
    CAST(NULL AS NUMBER)                    AS ma3,
    CAST(NULL AS VARCHAR2(12))              AS season_var,
    CAST(NULL AS NUMBER)                    AS net_avail,
    CAST(NULL AS VARCHAR2(18))              AS top_genre,
    CAST(NULL AS VARCHAR2(10))              AS festive_events,
    q.yr                                    AS ysort,
    q.qtr_num                               AS qsort,
    90                                      AS ord_in_qtr
  FROM q_stats q

  UNION ALL

  /* Quarter totals row AFTER the months (no Year here anymore) */
  SELECT
    CAST(NULL AS NUMBER)                    AS yr_display,
    'Q' || q.qtr_num || ' Total'            AS period_label,
    q.qtr_borrow_cnt                        AS borrows,
    q.prev_qtr                              AS last_period,
    q.pct_vs_last_qtr                       AS pct_vs_last,
    CAST(NULL AS NUMBER)                    AS ma3,
    CAST(NULL AS VARCHAR2(12))              AS season_var,
    q.net_avail_qtr                         AS net_avail,
    CAST(NULL AS VARCHAR2(18))              AS top_genre,
    CAST(NULL AS VARCHAR2(10))              AS festive_events,
    q.yr                                    AS ysort,
    q.qtr_num                               AS qsort,
    91                                      AS ord_in_qtr
```

```
FROM q_stats q

UNION ALL

/* Blank spacer AFTER quarter totals */
SELECT
  CAST(NULL AS NUMBER)                    AS yr_display,
  ' '                                     AS period_label,
  CAST(NULL AS NUMBER)                    AS borrows,
  CAST(NULL AS NUMBER)                    AS last_period,
  CAST(NULL AS NUMBER)                    AS pct_vs_last,
  CAST(NULL AS NUMBER)                    AS ma3,
  CAST(NULL AS VARCHAR2(12))              AS season_var,
  CAST(NULL AS NUMBER)                    AS net_avail,
  CAST(NULL AS VARCHAR2(18))              AS top_genre,
  CAST(NULL AS VARCHAR2(10))              AS festive_events,
  q.yr                                    AS ysort,
  q.qtr_num                               AS qsort,
  92                                      AS ord_in_qtr
FROM q_stats q

UNION ALL

/* Year subtotal band */
SELECT
  CAST(NULL AS NUMBER)                    AS yr_display,
  RPAD('*',12,'*')                        AS period_label,
  CAST(NULL AS NUMBER)                    AS borrows,
  CAST(NULL AS NUMBER)                    AS last_period,
  CAST(NULL AS NUMBER)                    AS pct_vs_last,
  CAST(NULL AS NUMBER)                    AS ma3,
  CAST(NULL AS VARCHAR2(12))              AS season_var,
  CAST(NULL AS NUMBER)                    AS net_avail,
  CAST(NULL AS VARCHAR2(18))              AS top_genre,
  CAST(NULL AS VARCHAR2(10))              AS festive_events,
  yt.yr                                   AS ysort,
  98                                      AS qsort,
  98                                      AS ord_in_qtr
FROM year_tot yt

UNION ALL
SELECT
  CAST(NULL AS NUMBER)                    AS yr_display,
  RPAD('Total',12,' ')                    AS period_label,
  yt.year_borrow_total                    AS borrows,
  CAST(NULL AS NUMBER)                    AS last_period,
  CAST(NULL AS NUMBER)                    AS pct_vs_last,
```

```
    CAST(NULL AS NUMBER)                    AS ma3,
    CAST(NULL AS VARCHAR2(12))              AS season_var,
    CAST(NULL AS NUMBER)                    AS net_avail,
    CAST(NULL AS VARCHAR2(18))              AS top_genre,
    CAST(NULL AS VARCHAR2(10))              AS festive_events,
    yt.yr                                   AS ysort,
    99                                      AS qsort,
    99                                      AS ord_in_qtr
  FROM year_tot yt

  UNION ALL

  /* Grand Total */
  SELECT
    CAST(NULL AS NUMBER)                    AS yr_display,
    RPAD('Grand Total',12,' ')              AS period_label,
    (SELECT SUM(year_borrow_total) FROM year_tot) AS borrows,
    CAST(NULL AS NUMBER)                    AS last_period,
    CAST(NULL AS NUMBER)                    AS pct_vs_last,
    CAST(NULL AS NUMBER)                    AS ma3,
    CAST(NULL AS VARCHAR2(12))              AS season_var,
    CAST(NULL AS NUMBER)                    AS net_avail,
    CAST(NULL AS VARCHAR2(18))              AS top_genre,
    CAST(NULL AS VARCHAR2(10))              AS festive_events,
    9999                                    AS ysort,
    99                                      AS qsort,
    999                                     AS ord_in_qtr
  FROM dual
)
ORDER BY ysort, qsort, ord_in_qtr;

TTITLE OFF
BTITLE OFF

PROMPT
PROMPT Legend:
PROMPT - Borrows = loan count.
PROMPT - Last Period = previous month / previous quarter.
PROMPT - Change % = vs Last.
PROMPT - 3-Month Moving Avg = trailing 3 months.
PROMPT - Seasonal Index % = month strength vs overall.
PROMPT - Net Copies Available = in-service book copies minus active loans.
PROMPT - Top Genre = most-borrowed genre for the month.
PROMPT - Holiday Month = Yes if any public holiday occurs in that month.
PROMPT

SPOOL OFF
```

Output:

```
Session altered.

=== Borrowing Trend Parameters ===
Enter START year (YYYY) [2023]: 2021
Enter END year (YYYY)   [2025]: 2023

RUN_DT
--------------------
06-SEP-2025
```

```
                          Quarterly and Monthly Borrowing Trend Analysis       2021 --       2023
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 Date: 06-SEP-2025

                        Borrows         Last   Change %         3-Month     Seasonal   Net Copies Top                 Holiday
 Year Period            (count)       Period    vs Last      Moving Avg      Index %    Available Genre                Month
----- ------------ ------------ ------------ ---------- --------------- ------------ ------------ ------------------ ----------
 2021 Q1    Jan            788                                   788.00     -1.92%         33,777 NON-FICTION        Yes
            Feb            749          788      -4.95          768.50     -8.56%         33,724 NON-FICTION        Yes
            Mar            869          749      16.02          802.00     +6.48%         33,632 NON-FICTION        No
      ------------
      Q1 Total           2,406                                                             33,632

      Q2    Apr            820          869      -5.64          812.67     -3.89%         33,566 NON-FICTION        No
            May            794          820      -3.17          827.67     +2.71%         33,554 NON-FICTION        Yes
            Jun            829          794       4.41          814.33     +1.40%         33,439 NON-FICTION        No
      ------------
      Q2 Total           2,443        2,406       1.54                                     33,439

      Q3    Jul            830          829       0.12          817.67     -1.76%         33,426 NON-FICTION        No
            Aug            822          830      -0.96          827.00     +3.90%         33,365 NON-FICTION        Yes
            Sep            832          822       1.22          828.00     +0.66%         33,264 NON-FICTION        Yes
      ------------
      Q3 Total           2,484        2,443       1.68                                     33,264

      Q4    Oct            784          832      -5.77          812.67     -1.35%         33,267 NON-FICTION        No
            Nov            824          784       5.10          813.33     +1.28%         33,207 NON-FICTION        No
            Dec            861          824       4.49          823.00     +1.03%         33,109 NON-FICTION        Yes
      ------------
      Q4 Total           2,469        2,484      -0.60                                     33,109

      ************
      Total              9,802
```

Quarterly and Monthly Borrowing Trend Analysis 2021 -- 2023

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Date: 06-SEP-2025

Year	Period		Borrows (count)	Last Period	Change % vs Last	3-Month Moving Avg	Seasonal Index %	Net Copies Available	Top Genre	Holiday Month
2022	Q1	Jan	771			771.00	-1.92%	33,101	NON-FICTION	Yes
		Feb	730	771	-5.32	750.50	-8.56%	33,034	NON-FICTION	Yes
		Mar	869	730	19.04	790.00	+6.48%	32,957	NON-FICTION	No
	Q1 Total		2,370					32,957		
	Q2	Apr	777	869	-10.59	792.00	-3.89%	32,910	NON-FICTION	No
		May	843	777	8.49	829.67	+2.71%	32,860	NON-FICTION	Yes
		Jun	820	843	-2.73	813.33	+1.40%	32,777	NON-FICTION	No
	Q2 Total		2,440	2,370	2.95			32,777		
	Q3	Jul	780	820	-4.88	814.33	-1.76%	32,761	NON-FICTION	No
		Aug	853	780	9.36	817.67	+3.90%	32,673	NON-FICTION	Yes
		Sep	830	853	-2.70	821.00	+0.66%	32,611	NON-FICTION	Yes
	Q3 Total		2,463	2,440	0.94			32,611		
	Q4	Oct	790	830	-4.82	824.33	-1.35%	32,582	NON-FICTION	No
		Nov	823	790	4.18	814.33	+1.28%	32,512	NON-FICTION	No
		Dec	820	823	-0.36	811.00	+1.03%	32,475	NON-FICTION	Yes
	Q4 Total		2,433	2,463	-1.22			32,475		

****

Quarterly and Monthly Borrowing Trend Analysis 2021 -- 2023

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Date: 06-SEP-2025

Year	Period		Borrows (count)	Last Period	Change % vs Last	3-Month Moving Avg	Seasonal Index %	Net Copies Available	Top Genre	Holiday Month
	Total		9,706							
2023	Q1	Jan	834			834.00	-1.92%	32,410	NON-FICTION	Yes
		Feb	752	834	-9.83	793.00	-8.56%	32,386	NON-FICTION	Yes
		Mar	860	752	14.36	815.33	+6.48%	32,286	NON-FICTION	No
	Q1 Total		2,446					32,286		
	Q2	Apr	748	860	-13.02	786.67	-3.89%	32,312	NON-FICTION	No
		May	869	748	16.18	825.67	+2.71%	32,198	NON-FICTION	Yes
		Jun	825	869	-5.06	814.00	+1.40%	32,109	NON-FICTION	No
	Q2 Total		2,442	2,446	-0.16			32,109		
	Q3	Jul	787	825	-4.61	827.00	-1.76%	32,101	NON-FICTION	No
		Aug	860	787	9.28	824.00	+3.90%	32,006	NON-FICTION	Yes
		Sep	794	860	-7.67	813.67	+0.66%	31,985	NON-FICTION	Yes
	Q3 Total		2,441	2,442	-0.04			31,985		
	Q4	Oct	833	794	4.91	829.00	-1.35%	31,915	NON-FICTION	No
		Nov	824	833	-1.08	817.00	+1.28%	31,850	NON-FICTION	No
		Dec	784	824	-4.85	813.67	+1.03%	31,834	NON-FICTION	Yes
	Q4 Total		2,441	2,441	0.00			31,834		

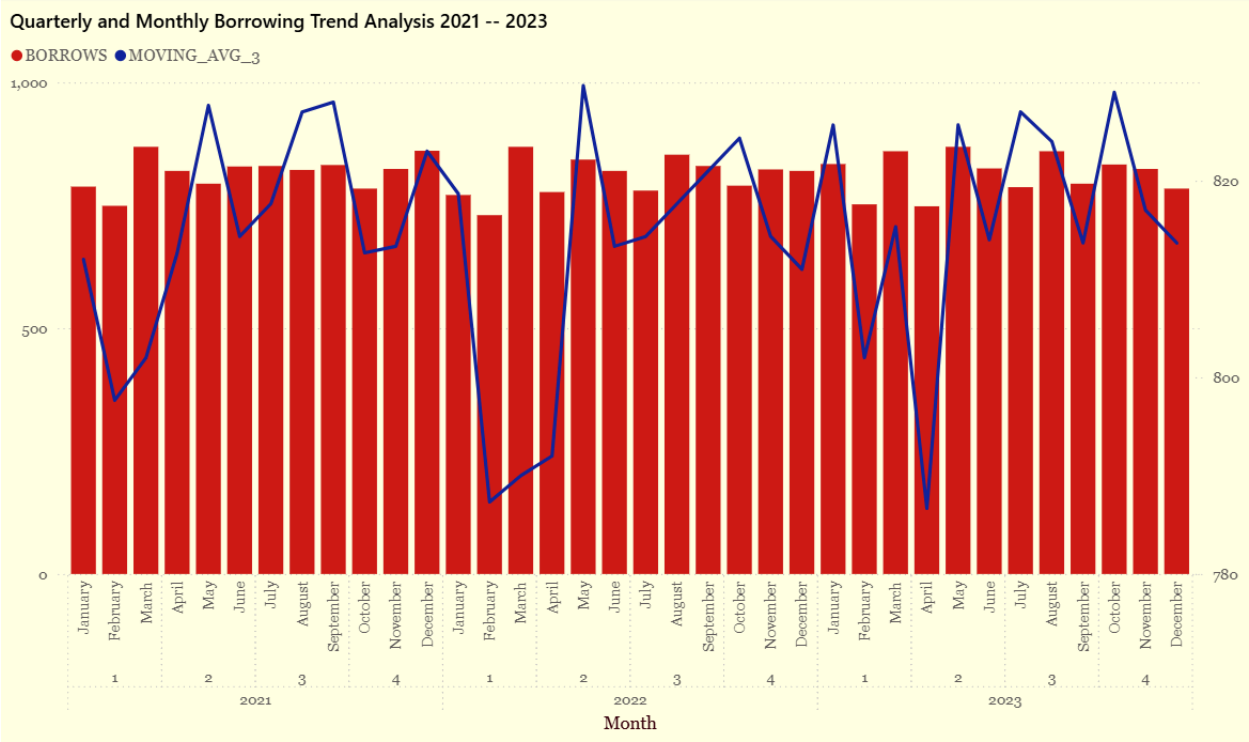
```
                          Quarterly and Monthly Borrowing Trend Analysis       2021 --       2023
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
Date: 06-SEP-2025

                        Borrows         Last   Change %         3-Month     Seasonal   Net Copies Top                 Holiday
 Year Period            (count)       Period    vs Last      Moving Avg      Index %    Available Genre                Month
----- ------------ ------------ ------------ ---------- --------------- ------------ ------------ ------------------ ----------
      ************
      Total               9,770

      Grand Total        29,278
```

```
79 rows selected.


Legend:
- Borrows = loan count.
- Last Period = previous month / previous quarter.
- Change % = vs Last.
- 3-Month Moving Avg = trailing 3 months.
- Seasonal Index % = month strength vs overall.
- Net Copies Available = in-service book copies minus active loans.
- Top Genre = most-borrowed genre for the month.
- Holiday Month = Yes if any public holiday occurs in that month.
```

The table presents a quarterly and monthly breakdown of library borrowing activity over the years 2021 to 2023, using data from the star schema (FactBorrowing joined to DimDate, and DimBook). It shows not just how many books were borrowed, but also how borrowing compares to previous periods, how seasonality and holidays affect patterns, and whether the library's book stock is sufficient to meet demand.

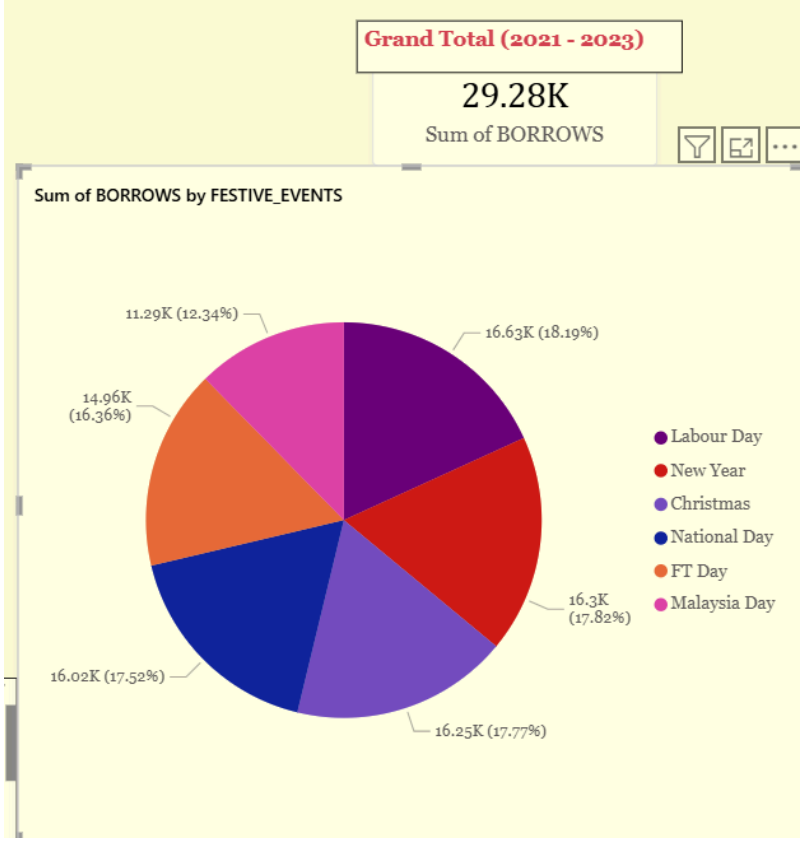
Between 2021 and 2023, borrowing stayed steady overall, with yearly totals moving only slightly up or down (about 9706 – 9802 loans each year). The real changes happened month to month, driven by short-term factors like holidays or COVID lockdowns, while quarterly volumes stayed in a narrow band. Meanwhile, net copies available fell from approximately 33.8k in January 2021 to 31.8k in December 2023 (≈ –6%). This gradual decline suggests collection retirement or slower replenishment of new titles, potentially increasing the risk of tighter stock during peak borrowing months.

From 2021 to 2023, borrowing was shaped by the pandemic and festive seasons. In 2021, numbers dipped in February (–4.95%) as MCO 2.0 limited movement and Chinese New Year activities kept people away, but activity later steadied. In 2022, April saw a sharp drop (–10.59%) during Ramadan and Hari Raya, when visits slowed due to fasting, family gatherings, and holiday closures, before rebounding in May (+8.49%) once routines resumed. The pattern was clearest in 2023, with February's dip (–9.83%) from the shorter month and extended CNY celebrations, and April's fall (–13.02%) again linked to Ramadan/Hari Raya, followed by a strong May recovery (+16.18%). Overall, the Seasonal Index confirms regular drops in February (≈ –8.56%) and April (≈ –3.89%) during major holidays, with rebounds in March, May, and August as borrowing picks up after festive breaks.

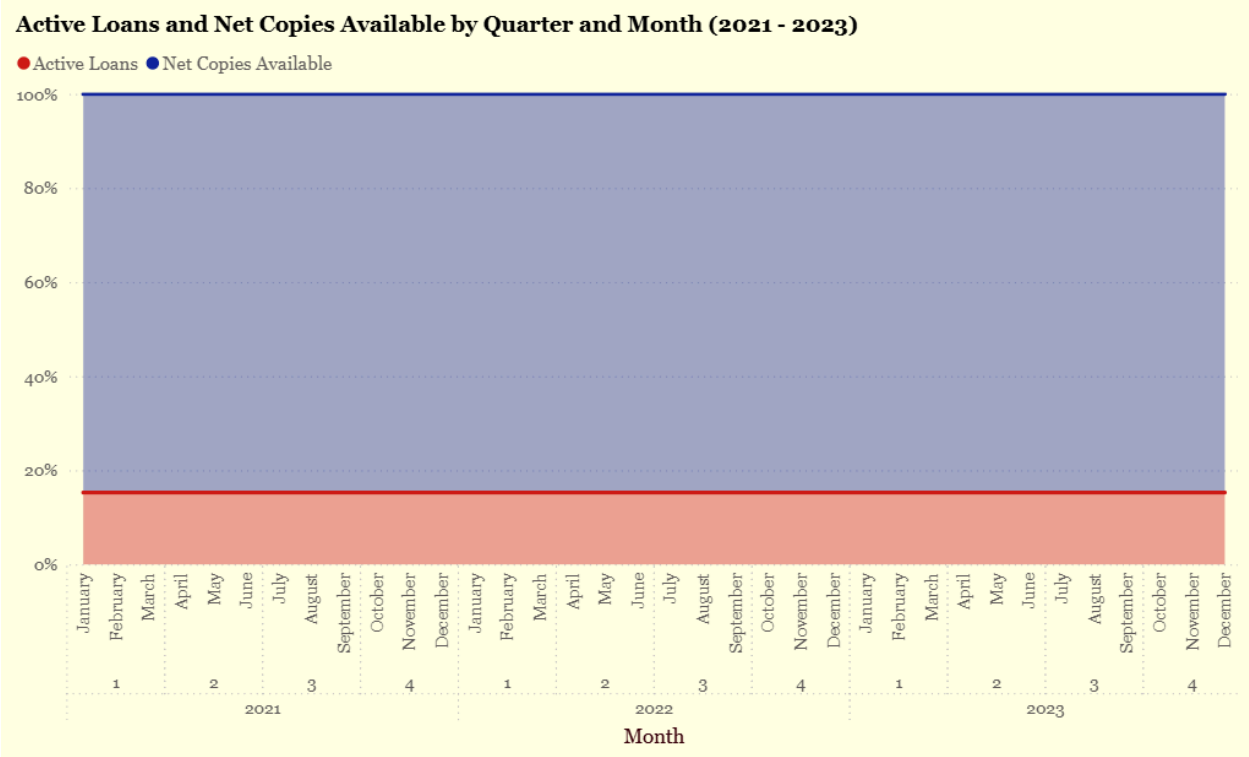


The **line and clustered column chart** shows **monthly borrowing** (red bars) and a **3-month moving average** (blue line) from **2021 to 2023**, helping to identify borrowing patterns, fluctuations, and overall stability in borrowing behavior over time.

Borrowing is generally stable but dips are visible in February and April, reflecting Chinese New Year, Ramadan, and Hari Raya periods when visits slow due to holidays, family activities, or closures. Sharp drops, such as April 2022 and April 2023, are followed by rebounds in May, highlighting a post-festive recovery effect. Occasional declines also appear in September (school holidays) and December (year-end breaks, weather disruptions). Overall, the pattern highlights predictable seasonal cycles rather than structural decline, offering clear opportunities for proactive library planning.



The **pie chart** shows **borrowing by festive events (2021–2023)**, totaling 29.28K borrows. Distribution is fairly balanced, with Labour Day (18.2%), New Year (17.8%), Christmas (17.8%), and National Day (17.5%) slightly higher. Malaysia Day records the smallest share (12.3%). This suggests that campaigns and extended loan offers could be prioritized around Labour Day, New Year, and Christmas to maximize impact, while Malaysia Day may need targeted promotions such as the "Read Malaysia" campaign to strengthen engagement and featuring local authors, history, and cultural books.



This chart shows the proportion of active loans versus net copies available over time to track library resource usage and availability. The chart shows active loans averaging 15–16% of total stock, with net copies consistently above 80%. This indicates sufficient capacity to meet demand. However, the gradual decline in book availability highlights the need for regular stock refresh and close monitoring of utilization to avoid future shortages during peak borrowing periods.

3.2.2 Sales and Borrowing Preference Analysis by Year and Genre

SQL Code:

```
SPOOL "C:\Users\wongj\Documents\BorrowVsSales_Genre_Report2.txt"

SET LINESIZE 130
SET PAGESIZE 35
SET TRIMSPOOL ON
SET VERIFY OFF
SET SQLBLANKLINES ON
SET WRAP OFF
SET TAB OFF
ALTER SESSION SET NLS_DATE_FORMAT = 'DD-MON-YYYY';

PROMPT === Borrow vs Sales Analysis Parameters ===
ACCEPT start_year NUMBER DEFAULT 2022 PROMPT 'Enter START year (YYYY) [2022]:
'
ACCEPT end_year   NUMBER DEFAULT 2025 PROMPT 'Enter END year (YYYY)   [2025]:
'

COLUMN run_dt NEW_VALUE run_dt
SELECT TO_CHAR(SYSDATE,'DD-MON-YYYY') run_dt FROM dual;

COLUMN sep_line NEW_VALUE sep_line NOPRINT
SELECT RPAD('_', 130, '_') AS sep_line FROM dual;

TTITLE LEFT ' ' CENTER 'Sales and Borrowing Preference Analysis by Year and
Genre &start_year -- &end_year' -
        SKIP 1 RIGHT 'Page ' FORMAT 999 SQL.PNO SKIP 1 LEFT 'Date: &run_dt'
SKIP 2
BTITLE CENTER '&sep_line'

-- ===== Column formats =====
CLEAR COLUMNS
COLUMN "Year"          HEADING 'Year'          FORMAT A17
COLUMN genre           HEADING 'Genre'         FORMAT A20
COLUMN borrow_cnt      HEADING 'Borrow|Count'  FORMAT 999,999
COLUMN sales_cnt       HEADING 'Sales|Count'   FORMAT 999,999
COLUMN borrow_pct      HEADING 'Borrow %'      FORMAT 990.00
COLUMN sales_pct       HEADING 'Sales %'       FORMAT 990.00
COLUMN revenue         HEADING 'Sales|Revenue' FORMAT 999,999,990.00
COLUMN male_pct        HEADING 'Member`s|Male %'        FORMAT 990.00
COLUMN female_pct      HEADING 'Gender|Female %'      FORMAT 990.00
COLUMN preference      HEADING 'Preference'    FORMAT A18

-- ===== Only show Year once per group; show grand totals at very end =====
CLEAR BREAKS
```

```
CLEAR COMPUTES
BREAK ON "Year" SKIP 1 ON REPORT
COMPUTE SUM LABEL 'Grand Total:' OF borrow_cnt sales_cnt revenue ON REPORT

-- ===== Base (Oracle 11g safe) =====
WITH
sales AS (
  SELECT d.cal_year,
         b.genre,
         COUNT(fs.sales_id)         AS sales_cnt,
         SUM(fs.line_total)         AS revenue,
          COUNT(DISTINCT CASE WHEN m.memberGender = 'M' THEN m.memberKey END)
AS male_sales,
          COUNT(DISTINCT CASE WHEN m.memberGender = 'F' THEN m.memberKey END)
AS female_sales
  FROM FactSales fs
  JOIN DimDate    d ON d.dateKey = fs.dateKey
  JOIN DimBook    b ON b.bookKey = fs.bookKey
  JOIN DimMembers m ON m.memberKey = fs.memberKey
  WHERE d.cal_year BETWEEN &start_year AND &end_year
  GROUP BY d.cal_year, b.genre
),
borrows AS (
  SELECT d.cal_year,
         b.genre,
         COUNT(*) AS borrow_cnt,
          COUNT(DISTINCT CASE WHEN m.memberGender = 'M' THEN m.memberKey END)
AS male_borrow,
          COUNT(DISTINCT CASE WHEN m.memberGender = 'F' THEN m.memberKey END)
AS female_borrow
  FROM FactBorrowing fb
  JOIN DimDate    d ON d.dateKey = fb.dateKey
  JOIN DimBook    b ON b.bookKey = fb.bookKey
  JOIN DimMembers m ON m.memberKey = fb.memberKey
  WHERE d.cal_year BETWEEN &start_year AND &end_year
  GROUP BY d.cal_year, b.genre
),
combined AS (
  SELECT NVL(s.cal_year, b.cal_year) AS cal_year,
         NVL(s.genre,    b.genre)    AS genre,
         NVL(b.borrow_cnt,0)         AS borrow_cnt,
         NVL(s.sales_cnt,0)          AS sales_cnt,
         NVL(s.revenue,0)            AS revenue,
         NVL(s.male_sales,0)  + NVL(b.male_borrow,0)    AS male_cnt,
         NVL(s.female_sales,0)+ NVL(b.female_borrow,0)  AS female_cnt
  FROM sales s
  FULL OUTER JOIN borrows b
```

```
    ON s.cal_year = b.cal_year
   AND s.genre    = b.genre
),
detail AS (
  SELECT
    cal_year,
    genre,
    borrow_cnt,
    sales_cnt,
    revenue,
       ROUND(borrow_cnt  /  NULLIF(borrow_cnt  +  sales_cnt,  0)  *  100,  2)  AS
borrow_pct,
       ROUND(sales_cnt   /  NULLIF(borrow_cnt  +  sales_cnt,  0)  *  100,  2)  AS
sales_pct,
       ROUND(male_cnt     /  NULLIF(male_cnt  +  female_cnt,  0)  *  100,  2)   AS
male_pct,
       ROUND(female_cnt  /  NULLIF(male_cnt  +  female_cnt,  0)  *  100,  2)   AS
female_pct,
    CASE
      WHEN borrow_cnt > sales_cnt * 1.10 THEN 'Borrowing Higher'
      WHEN sales_cnt  > borrow_cnt * 1.10 THEN 'Sales Higher'
      ELSE 'Balanced'
    END AS preference
  FROM combined
),
year_tot AS (   -- recompute % from totals (not averages)
  SELECT
    cal_year,
    'Total' AS genre,
    SUM(borrow_cnt) AS borrow_cnt,
    SUM(sales_cnt)  AS sales_cnt,
    SUM(revenue)    AS revenue,
    -- For demographics, recompute from distinct participants across the year.
    -- If you prefer distinct members across all genres for the year,
    -- switch to a separate subquery. Here we approximate from sums:
    SUM(male_cnt)   AS male_cnt,
    SUM(female_cnt) AS female_cnt
  FROM combined
  GROUP BY cal_year
),
year_tot_fmt AS (
  SELECT
    cal_year,
    '==================' AS genre,
    NULL AS borrow_cnt,
    NULL AS sales_cnt,
    NULL AS revenue,
```

```
    NULL AS borrow_pct,
    NULL AS sales_pct,
    NULL AS male_pct,
    NULL AS female_pct,
    NULL AS preference
  FROM year_tot
  UNION ALL
  SELECT
    cal_year,
    'Total' AS genre,
    borrow_cnt,
    sales_cnt,
    revenue,
       ROUND(borrow_cnt  /  NULLIF(borrow_cnt  +  sales_cnt,  0)  *  100,  2)  AS
borrow_pct,
       ROUND(sales_cnt   /  NULLIF(borrow_cnt  +  sales_cnt,  0)  *  100,  2)  AS
sales_pct,
       ROUND(male_cnt     /  NULLIF(male_cnt  +  female_cnt,  0)  *  100,  2)   AS
male_pct,
       ROUND(female_cnt  /  NULLIF(male_cnt  +  female_cnt,  0)  *  100,  2)   AS
female_pct,
    CASE
      WHEN borrow_cnt > sales_cnt * 1.10 THEN 'Borrowing Higher'
      WHEN sales_cnt  > borrow_cnt * 1.10 THEN 'Sales Higher'
      ELSE 'Balanced'
    END AS preference
  FROM year_tot
)
-- ===== Final output: details, then a "Total" row per year at bottom =====
SELECT *
FROM (
  SELECT
    TO_CHAR(d.cal_year) AS "Year",
    d.genre,
    d.borrow_cnt,
    d.sales_cnt,
    d.borrow_pct,
    d.sales_pct,
    d.revenue,
    d.male_pct,
    d.female_pct,
    d.preference
  FROM detail d

  UNION ALL

  SELECT
```

```
    TO_CHAR(t.cal_year) AS "Year",
    t.genre,
    t.borrow_cnt,
    t.sales_cnt,
    t.borrow_pct,
    t.sales_pct,
    t.revenue,
    t.male_pct,
    t.female_pct,
    t.preference
  FROM year_tot_fmt t
)
-- Order by: 1 = Year, put 'Total' after all genres in that year, then by
Genre
ORDER BY 1,
         CASE WHEN genre = '===================' THEN 1
              WHEN genre = 'Total'      THEN 2
              ELSE 0 END,
         2;

TTITLE OFF
BTITLE OFF

PROMPT
PROMPT Legend:
PROMPT • "Year" prints once; genres list under it. A "Total" row appears at
the end of each year.
PROMPT • Borrow Count = number of borrowing transactions; Sales Count = number
of sales transactions.
PROMPT • Sales Revenue = total sales amount (from line_total).
PROMPT • Borrow % = Borrow Count / (Borrow Count + Sales Count) × 100.
PROMPT • Sales  % = Sales Count  / (Borrow Count + Sales Count) × 100.
PROMPT • Gender % = share of distinct members (M/F) who borrowed or bought for
that line.
PROMPT
PROMPT • Preference (10% rule):
PROMPT   - Borrowing Higher: Borrow Count > 110% of Sales Count
PROMPT   - Sales Higher:     Sales Count  > 110% of Borrow Count
PROMPT   - Balanced:         Borrow/Sales between 0.91 and 1.10 (inclusive);
PROMPT
PROMPT • Preference Near-50% view (derived from 10% count gap):
PROMPT   - Borrowing Higher: Borrow % ≥ 52.38%
PROMPT   - Sales Higher:     Borrow % ≤ 47.62%
PROMPT   - Balanced:         Borrow % between 47.62% and 52.38% (inclusive);
PROMPT

SPOOL OFF
```

Output:

```
Session altered.

=== Borrow vs Sales Analysis Parameters ===
Enter START year (YYYY) [2022]: 2019
Enter END year (YYYY)   [2025]: 2023

RUN_DT
--------------------
07-SEP-2025
```

```
                    Sales and Borrowing Preference Analysis by Year and Genre       2019 --       2023
                                                                                                                    Page    1
Date: 07-SEP-2025

                                          Borrow    Sales                                Sales Member`s   Gender
Year              Genre                    Count    Count Borrow % Sales %          Revenue   Male % Female % Preference
----------------- -------------------- -------- -------- -------- ------- --------------- -------- -------- ------------------
2019              ADVENTURE                 775      647    54.50   45.50      123,707.19    36.90    63.10 Borrowing Higher
                  AUTOBIOGRAPHY             751      659    53.26   46.74      125,640.77    33.93    66.07 Borrowing Higher
                  FANTASY                   677      643    51.29   48.71      119,840.91    33.65    66.35 Balanced
                  FICTION                   692      694    49.93   50.07      139,688.13    34.98    65.02 Balanced
                  HISTORY                 1,312    1,401    48.36   51.64      274,431.11    34.86    65.14 Balanced
                  JUVENILE FICTION          707      647    52.22   47.78      121,884.38    35.29    64.71 Balanced
                  MYSTERY                   682      656    50.97   49.03      121,417.81    34.80    65.20 Balanced
                  NON-FICTION             2,054    2,053    50.01   49.99      387,777.52    34.27    65.73 Balanced
                  PHILOSOPHY                724      702    50.77   49.23      131,571.25    34.60    65.40 Balanced
                  ROMANCE                   739      681    52.04   47.96      130,548.18    35.16    64.84 Balanced
                  THRILLER                  669      664    50.19   49.81      133,624.14    33.97    66.03 Balanced
                  ===================
                  Total                   9,782    9,447    50.87   49.13    1,810,131.39    34.73    65.27 Balanced

2020              ADVENTURE                 727      646    52.95   47.05      121,776.11    35.72    64.28 Borrowing Higher
                  AUTOBIOGRAPHY             691      699    49.71   50.29      130,194.02    34.21    65.79 Balanced
                  FANTASY                   728      647    52.95   47.05      118,912.16    32.18    67.82 Borrowing Higher
                  FICTION                   725      664    52.20   47.80      121,916.79    37.16    62.84 Balanced
                  HISTORY                 1,370    1,345    50.46   49.54      248,901.28    37.05    62.95 Balanced
                  JUVENILE FICTION          723      699    50.84   49.16      129,088.91    34.51    65.49 Balanced
                  MYSTERY                   681      715    48.78   51.22      135,912.56    36.00    64.00 Balanced
                  NON-FICTION             2,111    2,036    50.90   49.10      380,714.71    34.13    65.87 Balanced
                  PHILOSOPHY                666      708    48.47   51.53      144,454.10    35.12    64.88 Balanced
                  ROMANCE                   708      619    53.35   46.65      114,480.56    34.62    65.38 Borrowing Higher
                  THRILLER                  676      689    49.52   50.48      131,901.02    36.76    63.24 Balanced
                  ===================
```

Sales and Borrowing Preference Analysis by Year and Genre 2019 -- 2023

Page 2

Date: 07-SEP-2025

Year	Genre	Borrow Count	Sales Count	Borrow %	Sales %	Sales Revenue	Member`s Male %	Gender Female %	Preference
2020	Total	9,806	9,467	50.88	49.12	1,778,252.22	35.21	64.79	Balanced
2021	ADVENTURE	723	712	50.38	49.62	143,300.27	34.14	65.86	Balanced
	AUTOBIOGRAPHY	751	702	51.69	48.31	128,620.10	32.97	67.03	Balanced
	FANTASY	744	740	50.13	49.87	136,872.75	34.70	65.30	Balanced
	FICTION	702	703	49.96	50.04	137,031.04	34.61	65.39	Balanced
	HISTORY	1,343	1,351	49.85	50.15	257,728.36	36.93	63.07	Balanced
	JUVENILE FICTION	679	720	48.53	51.47	133,413.70	33.79	66.21	Balanced
	MYSTERY	731	654	52.78	47.22	120,670.67	36.58	63.42	Borrowing Higher
	NON-FICTION	2,021	1,995	50.32	49.68	379,286.70	35.33	64.67	Balanced
	PHILOSOPHY	702	659	51.58	48.42	129,410.31	36.01	63.99	Balanced
	ROMANCE	719	674	51.62	48.38	125,255.41	33.66	66.34	Balanced
	THRILLER	687	693	49.78	50.22	131,492.03	36.26	63.74	Balanced
	Total	9,802	9,603	50.51	49.49	1,823,081.34	35.13	64.87	Balanced
2022	ADVENTURE	712	707	50.18	49.82	126,844.20	36.73	63.27	Balanced
	AUTOBIOGRAPHY	732	674	52.06	47.94	124,414.37	35.59	64.41	Balanced
	FANTASY	678	665	50.48	49.52	121,189.45	34.29	65.71	Balanced
	FICTION	691	682	50.33	49.67	128,869.95	37.33	62.67	Balanced
	HISTORY	1,392	1,319	51.35	48.65	241,310.72	34.19	65.81	Balanced
	JUVENILE FICTION	697	697	50.00	50.00	134,040.40	33.36	66.64	Balanced
	MYSTERY	689	717	49.00	51.00	126,712.55	33.78	66.22	Balanced
	NON-FICTION	2,002	2,105	48.75	51.25	375,992.54	36.57	63.43	Balanced
	PHILOSOPHY	712	704	50.28	49.72	132,201.20	34.35	65.65	Balanced
	ROMANCE	717	732	49.48	50.52	138,182.52	35.10	64.90	Balanced

Sales and Borrowing Preference Analysis by Year and Genre 2019 -- 2023

Page 3

Date: 07-SEP-2025

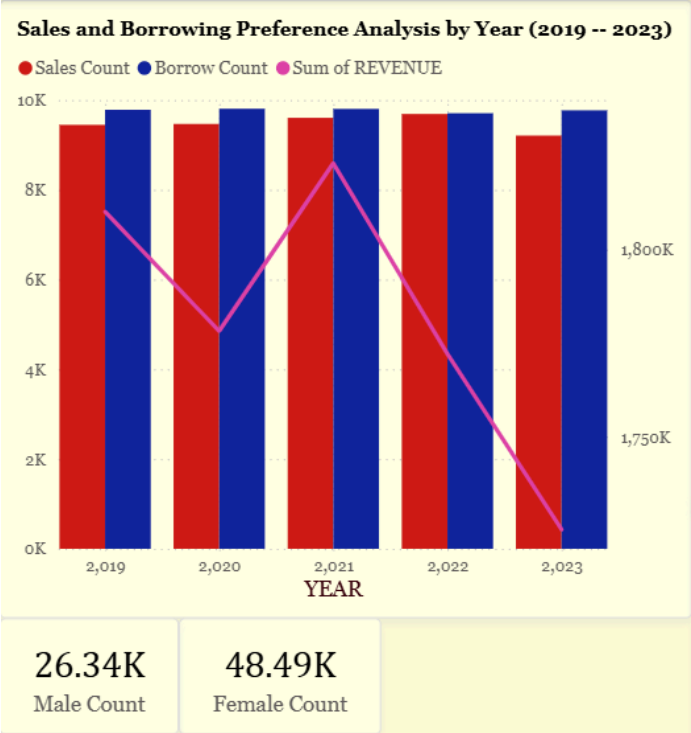
Year	Genre	Borrow Count	Sales Count	Borrow %	Sales %	Sales Revenue	Member`s Male %	Gender Female %	Preference
2022	THRILLER	684	687	49.89	50.11	122,327.73	34.73	65.27	Balanced
	Total	9,706	9,689	50.04	49.96	1,772,085.63	35.19	64.81	Balanced
2023	ADVENTURE	696	648	51.79	48.21	123,613.89	37.28	62.72	Balanced
	AUTOBIOGRAPHY	733	631	53.74	46.26	121,377.44	34.94	65.06	Borrowing Higher
	FANTASY	664	647	50.65	49.35	114,830.61	35.83	64.17	Balanced
	FICTION	743	681	52.18	47.82	130,228.51	34.54	65.46	Balanced
	HISTORY	1,428	1,369	51.05	48.95	257,603.42	35.55	64.45	Balanced
	JUVENILE FICTION	708	633	52.80	47.20	119,469.73	36.06	63.94	Borrowing Higher
	MYSTERY	686	627	52.25	47.75	117,273.54	34.43	65.57	Balanced
	NON-FICTION	2,057	1,934	51.54	48.46	360,027.28	34.90	65.10	Balanced
	PHILOSOPHY	695	667	51.03	48.97	122,727.86	36.10	63.90	Balanced
	ROMANCE	714	671	51.55	48.45	120,067.63	37.10	62.90	Balanced
	THRILLER	646	700	47.99	52.01	137,960.67	38.42	61.58	Balanced
	Total	9,770	9,208	51.48	48.52	1,725,180.58	35.78	64.22	Balanced
********** Grand Total:		97,732	94,828			17,817,462.32			

```
65 rows selected.


Legend:
• "Year" prints once; genres list under it. A "Total" row appears at the end of each year.
• Borrow Count = number of borrowing transactions; Sales Count = number of sales transactions.
• Sales Revenue = total sales amount (from line_total).
• Borrow % = Borrow Count / (Borrow Count + Sales Count) × 100.
• Sales  % = Sales Count  / (Borrow Count + Sales Count) × 100.
• Gender % = share of distinct members (M/F) who borrowed or bought for that line.

• Preference (10% rule):
– Borrowing Higher: Borrow Count > 110% of Sales Count
– Sales Higher:     Sales Count  > 110% of Borrow Count
– Balanced:         Borrow/Sales between 0.91 and 1.10 (inclusive)

• Preference Near-50% view (derived from 10% count gap):
– Borrowing Higher: Borrow % ≥ 52.38%
– Sales Higher:     Borrow % ≤ 47.62%
– Balanced:         Borrow % between 47.62% and 52.38% (inclusive)
```

The borrow-versus-sales analysis reveals that overall preferences remained **well-balanced** between 2019 and 2023, with borrowing averaging around 50–52% and sales 48–50%, suggesting members view the library as both a place to borrow and a retail-like outlet for book ownership. Genres such as non-fiction, fiction, fantasy, and history remain tightly balanced throughout, while some categories show stronger borrowing preferences in specific years: autobiography in 2022–2023, juvenile fiction in 2023, adventure in 2019–2020, and romance in 2020. These categories are often consumed quickly or have limited long-term use, so readers prefer borrowing rather than owning.

Revenue trends highlight the importance of sales despite near-equal transaction counts. For example, History and Non-Fiction consistently generated the highest revenue, aligning with their higher sales count. Gender participation remained stable across years, with **female members accounting for around 64–66%** of activity, pointing to strong engagement by women in both borrowing and purchasing.

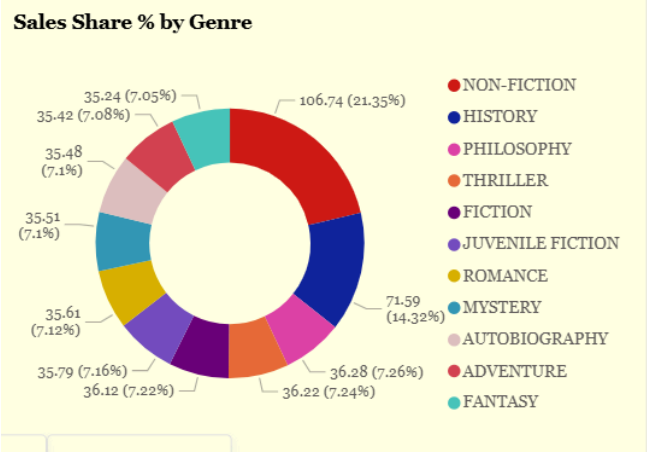
From a decision-making view, the library should focus on two things: keep enough books in popular genres for borrowing, and use sales to capture value in genres people like to own and reread. Promotions can help lift weaker areas (likeAutobiography or Mystery) and take advantage of seasonal peaks. Overall, borrowing and sales are both important, so the library needs to keep stock, prices, and marketing well balanced. Regularly reviewing borrowing and sales patterns will help the library adjust strategies quickly to changing member preferences.

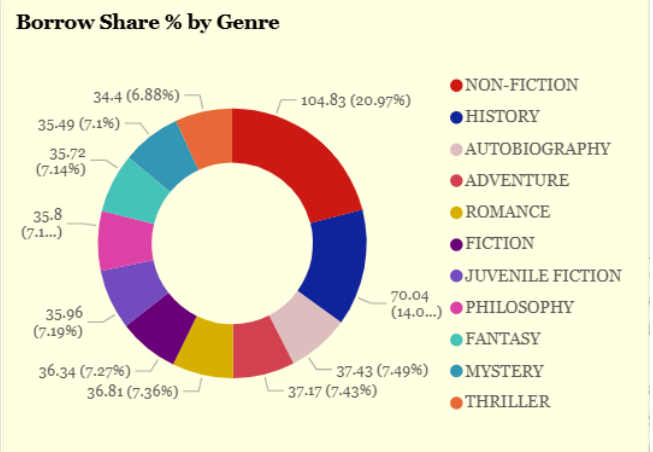


The line and clustered column chart compares **sales counts, borrow counts, and revenue (2019–2023)** to analyze trends in sales, borrowing behavior, and revenue, helping to identify customer preferences and their impact on financial performance. **Borrowing consistently exceeds sales** each year, confirming that members primarily rely on library access rather than purchases. Sales, however, remain close behind, showing the importance of maintaining both services. **Revenue peaks in 2021** as libraries and bookstores fully reopened after extended COVID-19 lockdowns in 2020, triggering a surge in both borrowing and book purchases as people sought learning and leisure materials. The **decline in 2022–2023** followed Malaysia's border reopening and economic recovery, as people shifted spending toward travel and entertainment while borrowing levels stayed steady.

Gender participation highlights a strong **female majority (≈65%)**, suggesting that promotional campaigns may benefit from targeting women's reading preferences while also encouraging greater male engagement.

From a decision-making view, the library should continue balancing borrowing and sales but also investigate why revenue dipped after 2021. This may involve adjusting pricing, refreshing popular saleable genres, or running bundled promotions to boost sales without reducing borrowing.





These donut charts show the **sales share and borrow share percentages by genre**, aiming to compare customer purchasing and borrowing preferences across different book categories.

The donut charts show that **Non-Fiction dominates both sales (21.35%) and borrowing (20.97%)**, followed by **History (14.32% sales; 14.01% borrowing)**, showing it is popular for both keeping and temporary reading. **Fiction and Juvenile Fiction** also hold strong shares, with Fiction slightly higher in both, due to its popularity among younger readers and parents who prefer to own children's books. **Autobiography** stays high in borrowings but lower in sales, because it usually offers one-time reading value, is often tied to current interest, and is easily available in libraries.

To act on these findings, the library should prioritize Non-Fiction and History acquisitions, ensuring enough stock for heavy demand. Meanwhile, for Autobiography, the focus should remain on borrowing services, but targeted marketing such as author talks or themed events could help raise purchase interest.

3.2.3 Supplier Contribution Deliveries by Geography

SQL Code:

```
SPOOL "C:\Users\wongj\Documents\SupplierPerformance_Report3.txt"

-- ======================== Session and Prompts =========================
SET LINESIZE 130
SET PAGESIZE 35
SET TRIMSPOOL ON
SET VERIFY OFF
SET SQLBLANKLINES ON
SET WRAP OFF
SET TAB OFF
ALTER SESSION SET NLS_DATE_FORMAT = 'DD-MON-YYYY';

PROMPT === Supplier City Geography Parameters ===
ACCEPT start_year  NUMBER DEFAULT 2023 PROMPT 'Enter START year (YYYY) [2023]:
'
ACCEPT end_year    NUMBER DEFAULT 2025 PROMPT 'Enter END year (YYYY)   [2025]:
'
ACCEPT period_view CHAR    DEFAULT 'Q'   PROMPT 'Period view [Q=Quarterly,
M=Monthly]: '

COLUMN run_dt NEW_VALUE run_dt
SELECT TO_CHAR(SYSDATE,'DD-MON-YYYY') run_dt FROM dual;

-- Dynamic label for contribution header: Q -> Qtr, M -> Mon (silent)
SET TERMOUT OFF
COLUMN perhdr NEW_VALUE perhdr NOPRINT
SELECT CASE WHEN UPPER('&period_view')='Q' THEN 'Qtr' ELSE 'Mon' END AS perhdr
FROM dual;
SET TERMOUT ON
SET TERMOUT OFF
COLUMN sep_line NEW_VALUE sep_line NOPRINT
SELECT RPAD('_', 130, '_') AS sep_line FROM dual;
SET TERMOUT ON

TTITLE LEFT ' ' CENTER 'Supplier Contribution Deliveries by Geography
&start_year -- &end_year' -
        SKIP 1 RIGHT 'Page ' FORMAT 999 SQL.PNO SKIP 1 LEFT 'Date: &run_dt'
SKIP 2
BTITLE CENTER '&sep_line'

-- ======================== Display formats ==========================
CLEAR COLUMNS
COLUMN "Year"         HEADING 'Year'                          FORMAT 9999
COLUMN "Period"       HEADING 'Quarter|Month'                 FORMAT A8
```

```
COLUMN "City"         HEADING 'Supplier|City'                  FORMAT A20
COLUMN "Orders"       HEADING 'Orders|(POs)'                   FORMAT 9,999
COLUMN "Value (RM)"    HEADING 'Purchases|Value (RM)'                FORMAT
999,999,999.99
COLUMN "Qty"            HEADING 'Units|Qty'                          FORMAT
999,999,999
COLUMN "Avg Order"     HEADING 'Avg|Order (RM)'                      FORMAT
999,999,999.99
COLUMN "Contrib %"      HEADING 'Contribution|% of &perhdr'          FORMAT
999,990.00
COLUMN "Priority"      HEADING 'Delivery|Priority'                 FORMAT A8
JUSTIFY RIGHT
COLUMN "Top Supplier" HEADING 'Top Supplier|(PurchaseValue)'   FORMAT A40

-- Show Year once; show Period once (blank line between periods)
CLEAR BREAKS
CLEAR COMPUTES
BREAK ON "Year" SKIP 1 ON "Period" SKIP 1 ON REPORT

-- ======================== Star-Schema Model ========================
WITH
-- Fact joined to dimensions, filtered by year range
f AS (
  SELECT
    dd.cal_year                           AS yr,
    TO_NUMBER(SUBSTR(dd.cal_quarter,2,1)) AS qtr_num,
    EXTRACT(MONTH FROM dd.cal_date)       AS mon_num,
    UPPER(TRIM(NVL(ds.city,'OTHERS')))    AS city,
    TRIM(ds.supplierName)                 AS supplier_name,
    fp.purchaseOrderId                    AS po_id,
    fp.totalAmount                        AS amount,
    fp.quantity                           AS qty
  FROM FactPurchase fp
  JOIN DimDate      dd ON dd.dateKey     = fp.dateKey
  JOIN DimSuppliers ds ON ds.supplierKey = fp.supplierKey
  WHERE dd.cal_year BETWEEN &start_year AND &end_year
),

-- Deduplicate order value correctly (header-vs-line totals)
f_enriched AS (
  SELECT
    yr, qtr_num, mon_num, city, supplier_name, po_id, qty,
    CASE
      WHEN ABS(MAX(amount) OVER (PARTITION BY po_id)
             - MIN(amount) OVER (PARTITION BY po_id)) <= 0.005
        THEN MAX(amount) OVER (PARTITION BY po_id)
      ELSE SUM(amount) OVER (PARTITION BY po_id)
```

```
    END AS order_value,
    ROW_NUMBER() OVER (PARTITION BY po_id ORDER BY po_id) AS rn_po
  FROM f
),

-- ================== Quarterly and Monthly city views =================
q_city AS (
  SELECT
    yr,
    'Q' || qtr_num                                AS period_label,
    qtr_num                                       AS period_ord,
    city,
    COUNT(DISTINCT po_id)                          AS total_orders,
    SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END) AS total_value,
    SUM(NVL(qty,0))                                AS total_qty,
    ROUND(
      SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END)
      / NULLIF(COUNT(DISTINCT po_id),0), 2
    ) AS avg_order_value
  FROM f_enriched
  GROUP BY yr, qtr_num, city
),
q_tot AS (
  SELECT yr, period_ord, SUM(total_value) AS period_value
  FROM q_city
  GROUP BY yr, period_ord
),
q_city_pct AS (
  SELECT
    q.*,
    ROUND(100 * q.total_value / NULLIF(t.period_value,0), 2) AS contrib_pct,
    CASE
      WHEN (100 * q.total_value / NULLIF(t.period_value,0)) >= 30 THEN 'HIGH'
      WHEN (100 * q.total_value / NULLIF(t.period_value,0)) >= 15 THEN 'MED'
      ELSE 'LOW'
    END AS priority_flag
  FROM q_city q
  JOIN q_tot  t ON t.yr = q.yr AND t.period_ord = q.period_ord
),

m_city AS (
  SELECT
    yr,
    TO_CHAR(ADD_MONTHS(DATE '2000-01-01', mon_num-1), 'Mon') AS period_label,
    mon_num                                       AS period_ord,
    city,
    COUNT(DISTINCT po_id)                          AS total_orders,
```

```
    SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END) AS total_value,
    SUM(NVL(qty,0))                                AS total_qty,
    ROUND(
      SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END)
      / NULLIF(COUNT(DISTINCT po_id),0), 2
    ) AS avg_order_value
  FROM f_enriched
  GROUP BY yr, mon_num, city
),
m_tot AS (
  SELECT yr, period_ord, SUM(total_value) AS period_value
  FROM m_city
  GROUP BY yr, period_ord
),
m_city_pct AS (
  SELECT
    m.*,
    ROUND(100 * m.total_value / NULLIF(t.period_value,0), 2) AS contrib_pct,
    CASE
      WHEN (100 * m.total_value / NULLIF(t.period_value,0)) >= 30 THEN 'HIGH'
      WHEN (100 * m.total_value / NULLIF(t.period_value,0)) >= 15 THEN 'MED'
      ELSE 'LOW'
    END AS priority_flag
  FROM m_city m
  JOIN m_tot  t ON t.yr = m.yr AND t.period_ord = m.period_ord
),

-- ============ Pick view and append *** + "Year Total:" ==============
pc_raw AS (
  SELECT * FROM q_city_pct WHERE UPPER('&period_view')='Q'
  UNION ALL
  SELECT * FROM m_city_pct WHERE UPPER('&period_view')='M'
),

/* ====== Supplier totals per period (to print Top Supplier + value) ====== */
m_sup AS (
  SELECT
    yr, mon_num AS period_ord,
    supplier_name,
     COUNT(DISTINCT po_id)                                                AS
sup_orders,
     SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END)                 AS
sup_value
  FROM f_enriched
  GROUP BY yr, mon_num, supplier_name
),
m_sup_top AS (
```

```
  SELECT yr, period_ord,
         supplier_name AS top_supplier,
         sup_value     AS top_value,
         ROW_NUMBER() OVER (
           PARTITION BY yr, period_ord
           ORDER BY sup_value DESC, sup_orders DESC, supplier_name
         ) AS rn
  FROM m_sup
),
q_sup AS (
  SELECT
    yr, qtr_num AS period_ord,
    supplier_name,
     COUNT(DISTINCT po_id)                                                AS
sup_orders,
     SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END)                 AS
sup_value
  FROM f_enriched
  GROUP BY yr, qtr_num, supplier_name
),
q_sup_top AS (
  SELECT yr, period_ord,
         supplier_name AS top_supplier,
         sup_value     AS top_value,
         ROW_NUMBER() OVER (
           PARTITION BY yr, period_ord
           ORDER BY sup_value DESC, sup_orders DESC, supplier_name
         ) AS rn
  FROM q_sup
),

/* Winner list respecting selected grain (Top Supplier per period) */
period_winner AS (
  SELECT yr, period_ord, top_supplier, top_value
  FROM q_sup_top
  WHERE UPPER('&period_view')='Q' AND rn = 1
  UNION ALL
  SELECT yr, period_ord, top_supplier, top_value
  FROM m_sup_top
  WHERE UPPER('&period_view')='M' AND rn = 1
),

/* ==== Year Champion by highest purchase value (not by wins) ==== */
sup_year AS (
  SELECT
    yr,
    supplier_name,
```

```
    COUNT(DISTINCT po_id) AS year_orders,
    SUM(CASE WHEN rn_po = 1 THEN order_value ELSE 0 END) AS year_value
  FROM f_enriched
  GROUP BY yr, supplier_name
),
year_top_pick AS (
  SELECT
    sy.yr,
    MAX(sy.supplier_name)
      KEEP (DENSE_RANK FIRST ORDER BY sy.year_value DESC, sy.year_orders DESC,
sy.supplier_name)
      AS top_supplier_year,
    MAX(sy.year_value)
      KEEP (DENSE_RANK FIRST ORDER BY sy.year_value DESC, sy.year_orders DESC,
sy.supplier_name)
      AS top_supplier_year_value
  FROM sup_year sy
  GROUP BY sy.yr
),

year_tot AS (
  SELECT
    yr,
    SUM(total_orders) AS orders_sum,
    SUM(total_value)  AS value_sum,
    SUM(total_qty)    AS qty_sum
  FROM pc_raw
  GROUP BY yr
),

pc AS (
  -- detail rows
  SELECT
    yr, period_ord, period_label, city,
       total_orders,  total_value,  total_qty,  avg_order_value,  contrib_pct,
priority_flag,
    1 AS sec_ord
  FROM pc_raw

  UNION ALL

  -- decorative separator BEFORE the Year Total row (***)
  SELECT
    yt.yr,
    998 AS period_ord,
        (SELECT  MAX(period_label)  FROM  pc_raw  r  WHERE  r.yr  =  yt.yr)  AS
period_label,
```

```
    RPAD('*', 16, '*') AS city,
    CAST(NULL AS NUMBER) AS total_orders,
    CAST(NULL AS NUMBER) AS total_value,
    CAST(NULL AS NUMBER) AS total_qty,
    CAST(NULL AS NUMBER) AS avg_order_value,
    CAST(NULL AS NUMBER) AS contrib_pct,
    CAST(NULL AS VARCHAR2(8)) AS priority_flag,
    2 AS sec_ord
  FROM year_tot yt

  UNION ALL

  -- Year Total row
  SELECT
    yt.yr,
    999 AS period_ord,
        (SELECT  MAX(period_label)  FROM  pc_raw  r  WHERE  r.yr  =  yt.yr)  AS
period_label,
    'Year Total:' AS city,
    yt.orders_sum, yt.value_sum, yt.qty_sum,
    CAST(NULL AS NUMBER)       AS avg_order_value,
    CAST(NULL AS NUMBER)       AS contrib_pct,
    CAST(NULL AS VARCHAR2(8))  AS priority_flag,
    3 AS sec_ord
  FROM year_tot yt
),

/* ---- Print winner once per period: mark the first city row in each period
---- */
pc2 AS (
  SELECT pc.*,
         ROW_NUMBER() OVER (
           PARTITION BY pc.yr, pc.period_ord
           ORDER BY pc.sec_ord, pc.city
         ) AS rn_in_period
  FROM pc
)

-- ========================= SINGLE OUTPUT =========================
SELECT
  pc2.yr                      AS "Year",
  pc2.period_label            AS "Period",
  pc2.city                    AS "City",
  pc2.total_orders            AS "Orders",
  pc2.total_value             AS "Value (RM)",
  pc2.total_qty               AS "Qty",
  pc2.avg_order_value         AS "Avg Order",
```

```
  pc2.contrib_pct             AS "Contrib %",
  LPAD(pc2.priority_flag, 8)  AS "Priority",
  CASE
    WHEN pc2.sec_ord = 1 AND pc2.rn_in_period = 1
      THEN pw.top_supplier
    WHEN pc2.sec_ord = 3
      THEN ytp.top_supplier_year
    ELSE NULL
  END AS "Top Supplier"
FROM pc2
LEFT JOIN period_winner pw
  ON pw.yr = pc2.yr AND pw.period_ord = pc2.period_ord
LEFT JOIN year_top_pick ytp
  ON ytp.yr = pc2.yr
ORDER BY pc2.yr, pc2.sec_ord, pc2.period_ord, pc2.city;

TTITLE OFF
BTITLE OFF

PROMPT
PROMPT Legend:
PROMPT • Grain = (Quarter/Month) by City within Year.
PROMPT • Orders = distinct purchase orders; Value (RM) = PO value after de-dup
at PO level.
PROMPT • Avg Order = Value ÷ Orders.
PROMPT • Delivery Priority (by Contrib % share):
PROMPT     HIGH  : ≥ 30%   (major share — prioritize/expedite)
PROMPT     MED   : 15–29.99%
PROMPT     LOW   : < 15%
PROMPT • "Top Supplier" (per &perhdr): supplier with the highest purchase
value in that period.
PROMPT • "Year Total:": Year Champion = supplier with the highest total
purchase value in that year.
PROMPT

SPOOL OFF
```

Output:

```
Session altered.

=== Supplier City Geography Parameters ===
Enter START year (YYYY) [2023]: 2019
Enter END year (YYYY)   [2025]: 2023
Period view [Q=Quarterly, M=Monthly]: Q

RUN_DT
--------------------
07-SEP-2025
```

```
                          Supplier Contribution Deliveries by Geography       2019 --       2023
                                                                                                                                  Page    1
Date: 07-SEP-2025

      Quarter  Supplier             Orders       Purchases        Units            Avg Contribution Delivery Top Supplier
 Year Month    City                  (POs)      Value (RM)          Qty      Order (RM)     % of Qtr Priority (PurchaseValue)
----- -------- -------------------- ------ --------------- ------------ --------------- ------------ -------- --------------------
 2019 Q1       KUALA LUMPUR              1       11,819.83          214       11,819.83        46.70     HIGH SASBADI HOLDINGS
               SHAH ALAM                 1       13,489.52          230       13,489.52        53.30     HIGH

      Q2       JOHOR BAHRU               3       42,363.64          623       14,121.21        20.61      MED PUSTAKA NASIONAL
               KOTA DAMANSARA            1       14,935.02          201       14,935.02         7.26      LOW
               KUALA LUMPUR              6       70,505.01        1,061       11,750.84        34.29     HIGH
               PETALING JAYA             5       62,672.71        1,078       12,534.54        30.48     HIGH
               SHAH ALAM                 1       15,114.58          170       15,114.58         7.35      LOW

      Q3       KOTA DAMANSARA            1       11,008.64          188       11,008.64         8.81      LOW SASBADI HOLDINGS
               KUALA LUMPUR              4       65,920.43          851       16,480.11        52.76     HIGH
               PETALING JAYA             1       10,991.02          208       10,991.02         8.80      LOW
               SHAH ALAM                 3       37,023.47          550       12,341.16        29.63      MED

      Q4       KOTA DAMANSARA            2       25,741.40          366       12,870.70        30.16     HIGH BOOKXCESS WHOLESALE
               KUALA LUMPUR              3       39,951.57          654       13,317.19        46.81     HIGH
               PETALING JAYA             2       19,651.90          319        9,825.95        23.03      MED
               ****************
               Year Total:              34      441,188.74        6,713                                         SASBADI HOLDINGS


 2020 Q1       GEORGE TOWN               1       14,072.53          220       14,072.53        19.66      MED SIMONANDSCHUSTER
               KUALA LUMPUR              3       51,687.42          686       17,229.14        72.20     HIGH
               PETALING JAYA             1        5,832.09          169        5,832.09         8.15      LOW

      Q2       GEORGE TOWN               1       10,328.77          183       10,328.77        16.15      MED MARSHALL CAVENDISH E
```

Supplier Contribution Deliveries by Geography 2019 -- 2023

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Date: 07-SEP-2025

Year	Quarter Month	Supplier City	Orders (POs)	Purchases Value (RM)	Units Qty	Avg Order (RM)	Contribution % of Qtr	Delivery Priority	Top Supplier (PurchaseValue)
2020	Q2	JOHOR BAHRU	1	10,974.62	177	10,974.62	17.16	MED	
		KUALA LUMPUR	1	19,709.63	255	19,709.63	30.81	HIGH	
		PETALING JAYA	2	22,954.38	320	11,477.19	35.88	HIGH	
	Q3	KOTA DAMANSARA	3	40,314.66	570	13,438.22	19.95	MED	MPH DISTRIBUTORS
		KUALA LUMPUR	7	89,380.59	1,398	12,768.66	44.23	HIGH	
		PETALING JAYA	5	72,386.95	1,116	14,477.39	35.82	HIGH	
	Q4	KOTA DAMANSARA	3	32,594.33	532	10,864.78	60.95	HIGH	GEMPAK STARZ
		KUALA LUMPUR	1	6,153.99	110	6,153.99	11.51	LOW	
		PETALING JAYA	1	14,729.24	184	14,729.24	27.54	MED	
		**********							
		Year Total:	30	391,119.20	5,920				MPH DISTRIBUTORS
2021	Q1	GEORGE TOWN	1	8,827.55	171	8,827.55	20.84	MED	PENGUIN RANDOM HOUSE
		KUALA LUMPUR	2	23,365.24	347	11,682.62	55.15	HIGH	
		PETALING JAYA	1	10,174.46	205	10,174.46	24.01	MED	
	Q2	KOTA DAMANSARA	1	10,230.88	144	10,230.88	44.66	HIGH	MARSHALL CAVENDISH E
		KUALA LUMPUR	1	12,677.36	139	12,677.36	55.34	HIGH	
	Q3	JOHOR BAHRU	1	12,280.40	250	12,280.40	12.95	LOW	MPH DISTRIBUTORS
		KUALA LUMPUR	3	31,274.76	583	10,424.92	32.99	HIGH	
		PETALING JAYA	3	39,371.08	523	13,123.69	41.53	HIGH	
		SHAH ALAM	1	11,878.22	205	11,878.22	12.53	LOW	

Supplier Contribution Deliveries by Geography 2019 -- 2023

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Date: 07-SEP-2025

Year	Quarter Month	Supplier City	Orders (POs)	Purchases Value (RM)	Units Qty	Avg Order (RM)	Contribution % of Qtr	Delivery Priority	Top Supplier (PurchaseValue)
2021	Q4	GEORGE TOWN	1	14,856.65	244	14,856.65	23.10	MED	PENGUIN RANDOM HOUSE
		JOHOR BAHRU	1	9,483.12	161	9,483.12	14.74	LOW	
		KUALA LUMPUR	2	30,192.06	458	15,096.03	46.94	HIGH	
		SHAH ALAM	1	9,793.76	143	9,793.76	15.23	MED	
		**********							
		Year Total:	19	224,405.54	3,573				PENGUIN RANDOM HOUSE
2022	Q1	GEORGE TOWN	1	7,480.82	160	7,480.82	14.07	LOW	PUSTAKA NASIONAL
		JOHOR BAHRU	1	17,381.97	230	17,381.97	32.70	HIGH	
		KUALA LUMPUR	1	15,069.67	259	15,069.67	28.35	MED	
		PETALING JAYA	1	13,228.38	206	13,228.38	24.88	MED	
	Q2	KOTA DAMANSARA	1	15,219.82	200	15,219.82	53.54	HIGH	GEMPAK STARZ
		PETALING JAYA	1	13,209.05	195	13,209.05	46.46	HIGH	
	Q3	KUALA LUMPUR	1	4,525.59	106	4,525.59	9.22	LOW	TIMES PUBLISHING
		PETALING JAYA	2	28,937.29	389	14,468.65	58.96	HIGH	
		SHAH ALAM	1	15,613.99	220	15,613.99	31.82	HIGH	
	Q4	KOTA DAMANSARA	1	14,190.83	240	14,190.83	51.73	HIGH	BOOKXCESS WHOLESALE
		KUALA LUMPUR	1	13,242.21	195	13,242.21	48.27	HIGH	
		**********							
		Year Total:	12	158,099.62	2,400				OXFORD FAJAR

Supplier Contribution Deliveries by Geography 2019 -- 2023

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Date: 07-SEP-2025

Year	Quarter Month	Supplier City	Orders (POs)	Purchases Value (RM)	Units Qty	Avg Order (RM)	Contribution % of Qtr	Delivery Priority	Top Supplier (PurchaseValue)
2023	Q1	GEORGE TOWN	2	24,446.16	333	12,223.08	42.80	HIGH	PENGUIN RANDOM HOUSE
		KUALA LUMPUR	3	32,670.58	477	10,890.19	57.20	HIGH	
	Q2	KUALA LUMPUR	5	75,850.34	966	15,170.07	87.66	HIGH	MACMILLAN PUBLISHERS
		PETALING JAYA	1	10,678.16	237	10,678.16	12.34	LOW	
	Q3	GEORGE TOWN	1	15,238.98	200	15,238.98	19.09	MED	ITBM MALAYSIA
		KUALA LUMPUR	4	46,140.77	721	11,535.19	57.81	HIGH	
		SHAH ALAM	1	18,435.94	251	18,435.94	23.10	MED	
	Q4	GEORGE TOWN	2	35,876.56	446	17,938.28	29.36	MED	SIMONANDSCHUSTER
		KUALA LUMPUR	6	86,303.77	1,239	14,383.96	70.64	HIGH	
		**********							
		Year Total:	25	345,641.26	4,870				MACMILLAN PUBLISHERS

70 rows selected.

Legend:
- Grain = (Quarter/Month) by City within Year.
- Orders = distinct purchase orders; Value (RM) = PO value after de-dup at PO level.
- Avg Order = Value ÷ Orders.
- Delivery Priority (by Contrib % share):

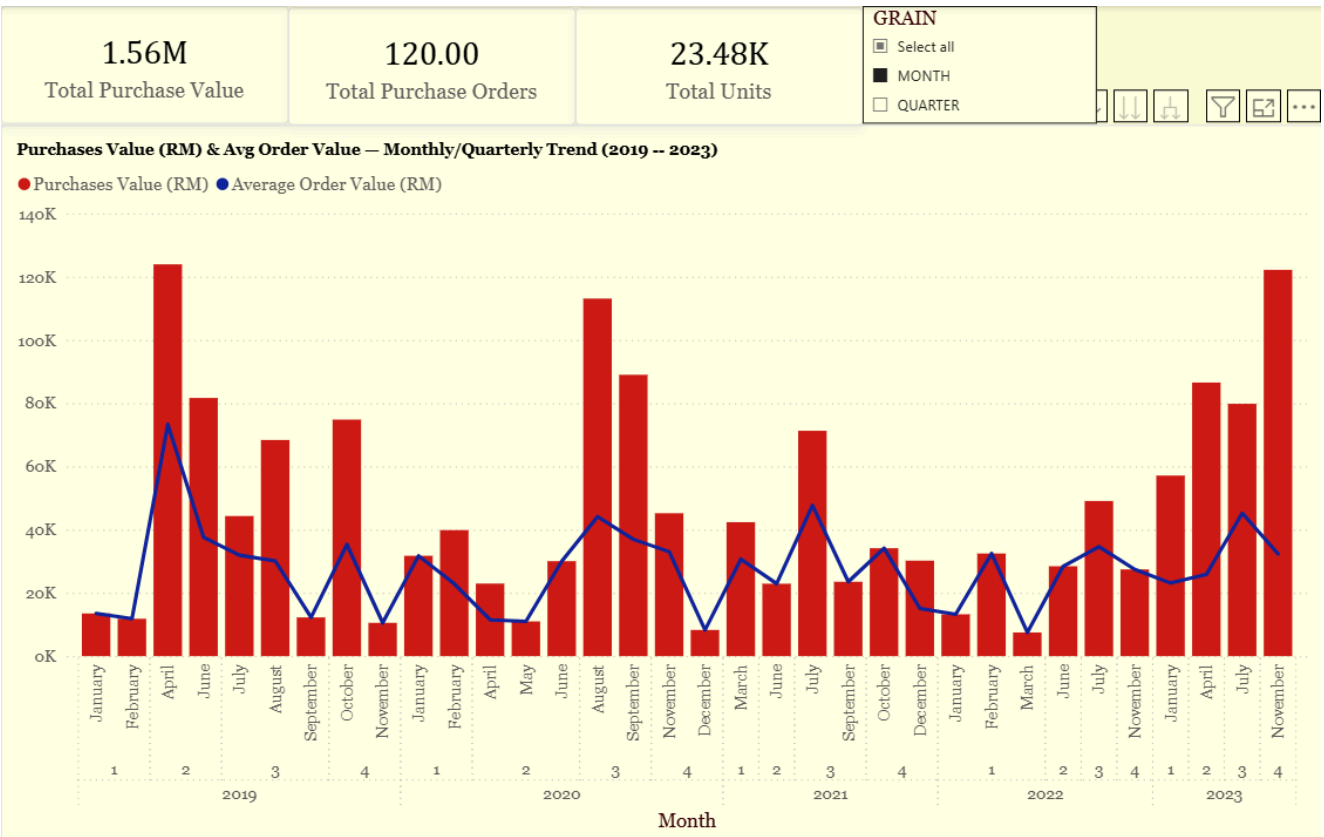
HIGH : ≥ 30% (major share — prioritize/expedite)
MED : 15–29.99%
LOW : < 15%

- "Top Supplier" (per Qtr): supplier with the highest purchase value in that period.
- "Year Total:": Year Champion = supplier with the highest total purchase value in that year.

The report shows supplier contribution analysis across 2019 to 2023. It was found that purchasing was **city-concentrated around Kuala Lumpur (KL)** in most quarters, with the share spiking to **~88% in 2023 Q2** and **~71% in 2023 Q4**. In contrast, **2019** displayed a more diversified pattern, with significant contributions also coming from Petaling Jaya, Shah Alam, and Johor Bahru. The contrast may be because purchasing strategies and supplier partnerships may have gradually centralized in Kuala Lumpur over time due to better logistics, and cost efficiencies, whereas in 2019 purchases were spread across several cities to balance availability and demand.

Purchases fell from **RM441k (2019)** to **RM158k (2022)**, then **rebounded to RM346k (2023),** supported by easier imports and faster fulfilment from Klang Valley suppliers, with Macmillan Publishers being the top supplier**.** The sharp decline from 2019 to 2022 was mainly due to the COVID-19 pandemic, which caused lockdowns, supply chain issues, and reduced demand as schools and libraries operated remotely. The rebound in 2023 reflects Malaysia's economic recovery and the reopening of education and retail sectors, though still below 2019 levels, as many readers shifted to online platforms following the widespread adoption of digital learning during the pandemic.

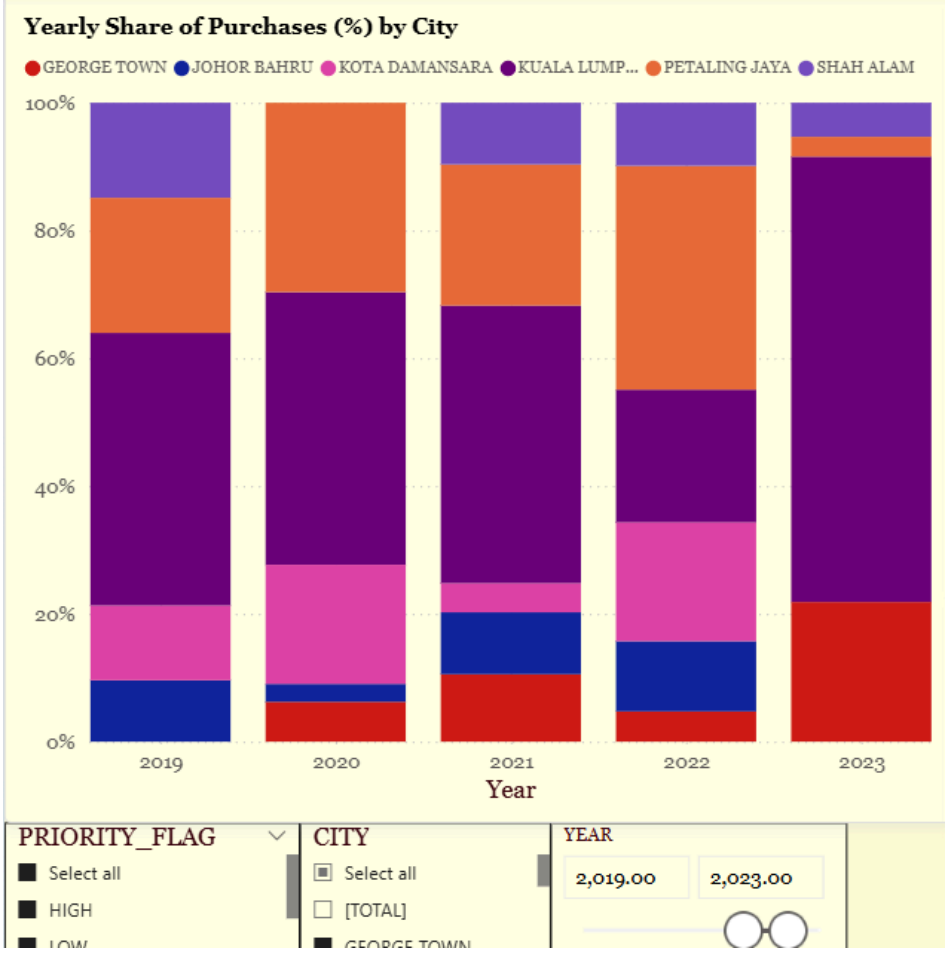
From a decision-making view, the concentration in Kuala Lumpur suggests the need to strengthen supplier relationships and logistics capacity there, while also diversifying sourcing to reduce dependency risk. At the same time, monitoring digital reading trends and adjusting procurement mix between physical and digital materials will be critical for long-term sustainability.



The chart shows **monthly purchase values and average order values** over time to identify spending trends, seasonal patterns, and shifts in customer purchasing behavior, highlighting a clear purchase cycle from **2019 to 2023**, with total purchase value (red bars) and average order value (blue line) moving in parallel. Purchases peaked strongly in early 2019, with April reaching about RM124k, reflecting a period of high-volume, high-value buying. After this, order values stabilized, though spending remained relatively strong across mid-year months such as June, August, and October.

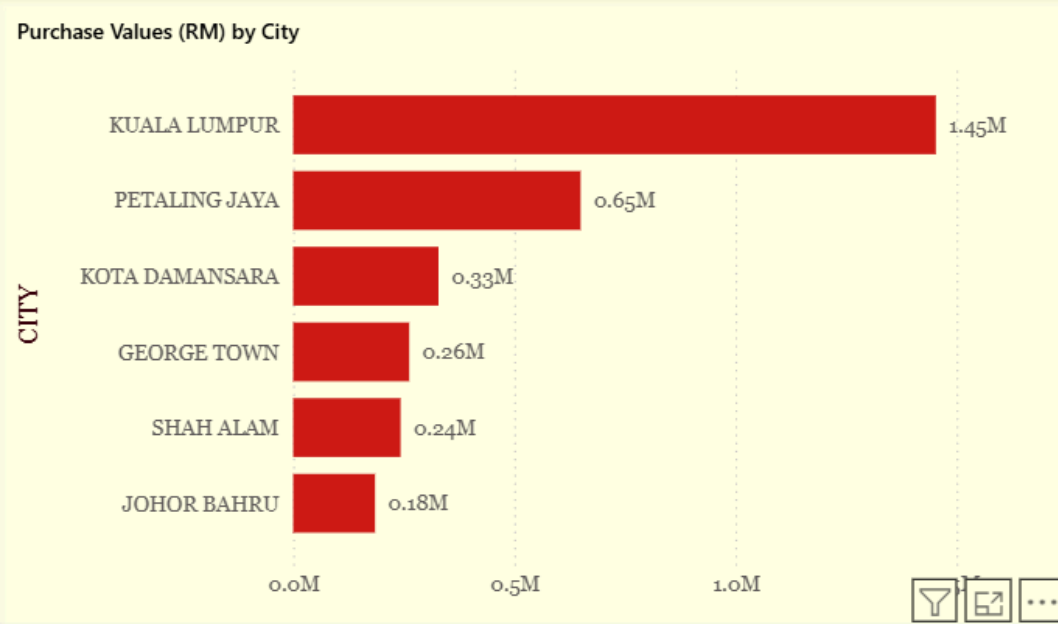
From 2020 to 2022, the impact of the COVID-19 pandemic is visible. Purchases became volatile in 2020, with a spike in August followed by lower and fragmented activity in 2021. The weakest year was 2022, when monthly purchase values mostly stayed between RM20k–40k and average order sizes dipped sharply, consistent with restricted budgets and reduced demand from schools and libraries.

In 2023, a strong recovery is evident. Purchases rose steadily from Q2 onwards, culminating in a November peak close to 2019 highs. However, average order values, though stronger than in 2022, remain below 2019 levels, suggesting more cautious purchasing. This highlights the need for careful cash flow planning around seasonal peaks and a balanced sourcing strategy to manage both bulk and steady demand.



The bar chart illustrates **yearly purchase shares by city from 2019 to 2023**, showing how supplier contributions shifted across hubs. Kuala Lumpur consistently dominates and expands further by 2023, signaling its role as the primary purchasing center. Petaling Jaya held significant shares until 2022 but dropped sharply in 2023, while Shah Alam, Johor Bahru, and Kota Damansara show smaller, fluctuating contributions. George Town, however, strengthened in the final year, turning as a growing alternative hub. This pattern reveals increasing dependence on Kuala Lumpur, which, while efficient, increases supply-chain risk.

From a decision-making perspective, the library should facilitate Kuala Lumpur's strong supply base but actively diversify purchasing toward rising hubs such as George Town and re-engage with Petaling Jaya. This will balance supply resilience, reduce risk from over-concentration, and position the library to benefit from competitive pricing across multiple cities. At the same time, establishing backup delivery routes and regional warehouses outside Kuala Lumpur will ensure supplies continue moving smoothly if the main hub faces delays.



The overall report already highlights the increasing centralisation of procurement in Kuala Lumpur. The **purchase value by city chart** reinforces this finding, showing KL as the dominant hub with RM1.45M, far ahead of Petaling Jaya (RM0.65M), which are useful for identifying key geographic markets. Other cities such as Kota Damansara, George Town, Shah Alam, and Johor Bahru contribute much smaller shares.